Writing an adapter

Developer quick-sheet · for anyone who wants goldpan to read another file format · version 2, 6 November 2026

The full guide is now part of the goldpan documentation: https://villano-lab.github.io/goldpan/guides/writing-an-adapter.html. It has the rules, the functions, the tests and the steps to make an adapter available, with the code. This sheet is the overview.

An **adapter** is the only code that knows a file format. It turns one storage (a MIDAS file, an idx dump, a ROOT file, a format of your own) into the objects the rest of `goldpan` works on; everything else in the library sees arrays and never a file. A new one is a small Python module, and the guide starts from a complete, tested example of about 50 lines.

What to know before you start

- **Two shapes.** An *events adapter* reads a file with waveforms into `goldpan.data.Events`; a *table adapter* reads processed quantities into a table that `goldpan.adapters.rq.attach` joins to an `Events`.
- **The user names the detector, the channels and the readout**, and the adapter checks them against what the file knows. A silent default would change every time and energy without an error.
- **The key of an event is `series`, `dump`, `event`**, because event numbers restart in every dump.
- **Skip and count, never invent**; raise a clear error for a feature you do not read yet.
- **Test it** with `goldpan.testing.check_adapter`, then register it (`goldpan.adapters.register`) for your own use, or add it to goldpan with a pull request.

Where things are

You want	Go to
the guide, step by step	https://villano-lab.github.io/goldpan/guides/writing-an-adapter.html
the example adapter	`tests/example_adapter.py` in https://github.com/villano-lab/goldpan
what the existing adapters offer	the reference of the goldpan documentation, https://villano-lab.github.io/goldpan/reference/index.html
how to contribute it	`CONTRIBUTING.md` in the goldpan repository

A new dependency of an adapter is named as an extra in goldpan and, when students need it, pinned through the `goldpan[...]` line of `requirements.txt` of this repository, with a *Students, please:* line at the next release (`RELEASING.md`).